Analytic Affine-Conditioned $K = 2$ Propagation for Coordinate-Spiked ReLU Networks

A rigorous continuous-slice formulation, finite-grid algorithm, and error decomposition

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July 2026

Abstract

We formalize an analytic alternative to binned K-propagation for a ReLU multilayer perceptron whose weight matrices contain a coordinate spike along e_1. The spike coordinate is propagated by a one-dimensional law, while the orthogonal bulk is represented conditionally by its first two moments. The central modeling choice is temporal: after a ReLU, the conditional bulk mean and covariance are retained as generally nonlinear functions of the nonnegative spike value; only after the next linear transformation and Bayesian reconditioning onto the new spike coordinate are these functions projected back onto the affine family

$$\mu(y) = \mu_0 + \mu_1 y, \qquad \Sigma(y) = \Sigma_0 + \Sigma_1 y.$$

The Gaussian–ReLU moment map is exact at each affine preactivation slice. Negative spike preactivations are collapsed into the genuine ReLU atom at zero by the laws of total expectation and total covariance. We derive the ideal continuous operator, an implementable finite-grid version with closed-form truncated-Gaussian updates, the best affine projection formulas, a covariance-preserving correction, and the exact error budget. Apart from optional one-dimensional discretization, the only conceptual approximations are conditional $K = 2$ Gaussian closure and the affine re-projection after linear reconditioning.

Contents

1 Network, spike–bulk split, and design principle

Let f be a depth-L ReLU multilayer perceptron with hidden width n,

$$Z^{(\ell)} = M^{(\ell)} X^{(\ell-1)} + b^{(\ell)}, \qquad X^{(\ell)} = \rho(Z^{(\ell)}), \qquad \rho(t) = \max\{t, 0\}, \tag{1}$$

where ReLU is applied coordinatewise. The motivating model has

$$M^{(\ell)} = W^{(\ell)} + e_1 e_1^\top, \qquad W_{ij}^{(\ell)} \sim \mathcal{N}(0, 1/n), \tag{2}$$

but the propagation formulas below apply to any deterministic realization of the matrices.

Write $e = e_1$, $\Pi = I - ee^\top$, and $d = n - 1$. Every layer vector is decomposed as

$$X = Ae + B, \qquad A = e^\top X \in \mathbb{R}, \qquad B = \Pi X \in e^\perp \cong \mathbb{R}^d. \tag{3}$$

Relative to $\mathbb{R}^n = \mathbb{R}e \oplus e^\perp$, a matrix and bias have the block form

$$M = \begin{pmatrix} \gamma & r^\top \\ u & V \end{pmatrix}, \qquad b = \begin{pmatrix} \beta \\ \eta \end{pmatrix}. \tag{4}$$

Thus a linear layer sends

$$Ae + B \longmapsto Ye + C, \qquad Y = \gamma A + r^\top B + \beta, \qquad C = uA + VB + \eta. \tag{5}$$

The algorithm alternates two different state geometries.

(i) After ReLU: the spike is nonnegative and has a genuine atom at zero. The conditional bulk mean and covariance are allowed to be nonlinear functions of the spike value.

(ii) Immediately before ReLU: the conditional bulk law is re-approximated by a Gaussian whose mean and covariance depend affinely on the signed spike preactivation.

The affine "re-pretend" therefore occurs only after a linear transformation has mixed the old spike and bulk and after we have reconditioned on the new spike coordinate. It does *not* occur immediately after ReLU.

2 State spaces and closure conventions

2.1 Post-ReLU nonlinear-slice state

Definition 2.1 (Nonlinear-slice $K = 2$ state)**.** A nonlinear-slice state is a triple

$$\mathsf{S}^+ = (\nu, m, S), \tag{6}$$

where

- $\nu = \mathcal{L}(A)$ is a probability measure on a measurable scalar domain $\mathcal{A} \subseteq \mathbb{R}$;
- $m(a) \in \mathbb{R}^d$ represents $\mathbb{E}[B \mid A = a]$;
- $S(a) \in \mathbb{S}_+^d$ represents $\mathrm{Cov}(B \mid A = a)$.

The $K = 2$ closure used by the next linear layer is

$$B \mid A = a \quad \rightsquigarrow \quad \mathcal{N}(m(a), S(a)). \tag{7}$$

No affine dependence on a is assumed in this state. After a ReLU layer, $\mathcal{A} \subseteq [0, \infty)$ and the law generally has an atom at 0.

The notation in (7) is deliberately an approximation arrow. Even when the incoming preactivation slice is Gaussian, applying ReLU makes the resulting bulk slice non-Gaussian. We compute its first two moments exactly and discard its conditional cumulants of order at least three when the next linear layer begins.

A practical finite representation is

$$\nu \approx \sum_{i=0}^{m-1} p_i \delta_{a_i}, \qquad a_i \geq 0, \qquad \sum_i p_i = 1, \tag{8}$$

with arrays

$$m_i = m(a_i), \qquad S_i = S(a_i). \tag{9}$$

The index $i = 0$ is reserved for $a_0 = 0$ when the zero atom is present.

2.2 Affine preactivation state

Definition 2.2 (Affine preactivation $K = 2$ state)**.** An affine preactivation state consists of a signed scalar law $\nu_Y = \mathcal{L}(Y)$ and coefficients

$$\mu_0, \mu_1 \in \mathbb{R}^d, \qquad \Sigma_0, \Sigma_1 \in \mathrm{Sym}_d, \tag{10}$$

with the conditional Gaussian model

$$C \mid Y = y \quad \rightsquigarrow \quad \mathcal{N}(\mu(y), \Sigma(y)), \qquad \mu(y) = \mu_0 + \mu_1 y, \quad \Sigma(y) = \Sigma_0 + \Sigma_1 y. \tag{11}$$

In a finite-grid implementation, positive semidefiniteness is required at every retained scalar node.

The ideal continuous formulas will first produce generally nonlinear conditional functions

$$\widehat{m}(y) = \mathbb{E}[C \mid Y = y], \qquad \widehat{S}(y) = \mathrm{Cov}(C \mid Y = y), \tag{12}$$

under the current $K = 2$ surrogate. The affine coefficients in (11) are then obtained by a precisely specified projection.

2.3 Two scalar-law backends

The bulk algorithm is independent of how the one-dimensional spike law is stored.

Continuous mixture backend. After a linear layer, the exact scalar law under the current conditional-Gaussian surrogate is a one-dimensional Gaussian mixture. It can be retained symbolically and integrated by adaptive one-dimensional quadrature. This removes conceptual spike quantization error, but a nonconstant globally affine covariance cannot remain positive semidefinite on all of $\mathbb{R}$; see theorem 6.3. A continuous implementation must therefore use a bounded fitting interval, a clipped scalar feature, or a PSD parameterization.

Finite-grid backend. The scalar law is represented by cells or quadrature nodes. The affine covariance is required to be PSD only at those nodes. This is the recommended implementation because it preserves the intended notation $\Sigma_0 + \Sigma_1 y$ without a global domain inconsistency, and because all negative nodes can be merged exactly into the ReLU zero atom.

3 Exact Gaussian–ReLU first and second moments

The ReLU step needs the first two moments of a coordinatewise rectified Gaussian.

Definition 3.1 (Gaussian–ReLU moment map)**.** For $G \sim \mathcal{N}(m, K)$ in $\mathbb{R}^d$, define

$$\mathfrak{m}_\rho(m, K) = \mathbb{E}[\rho(G)] \in \mathbb{R}^d, \tag{13}$$

$$\mathfrak{Q}_\rho(m, K) = \mathbb{E}[\rho(G)\rho(G)^\top] \in \mathrm{Sym}_d, \tag{14}$$

and

$$\mathfrak{S}_\rho(m, K) = \mathfrak{Q}_\rho(m, K) - \mathfrak{m}_\rho(m, K)\mathfrak{m}_\rho(m, K)^\top. \tag{15}$$

For one coordinate, let $G_i \sim \mathcal{N}(m_i, s_i^2)$ and $z_i = m_i/s_i$. When $s_i > 0$,

$$[\mathfrak{m}_\rho(m, K)]_i = m_i\Phi(z_i) + s_i\phi(z_i), \tag{16}$$

$$[\mathfrak{Q}_\rho(m, K)]_{ii} = (m_i^2 + s_i^2)\Phi(z_i) + m_i s_i \phi(z_i). \tag{17}$$

The degenerate case $s_i = 0$ is obtained by continuity and equals $\rho(m_i)$ and $\rho(m_i)^2$.

For completeness, every off-diagonal entry also has an exact analytic representation in terms of the bivariate Gaussian CDF. Let $I = (i, j)$, let $m_I \in \mathbb{R}^2$ and $K_I \in \mathbb{S}^2_+$ be the corresponding subvector and principal submatrix, let

$$D_I = \mathrm{diag}(\sqrt{(K_I)_{11}}, \sqrt{(K_I)_{22}}), \qquad R_I = D_I^{-1} K_I D_I^{-1}, \tag{18}$$

and let $\Phi_2(\cdot; R_I)$ be the centered bivariate Gaussian CDF with correlation matrix R_I. Define

$$\Psi_I(t) = \exp\left(t^\top m_I + \tfrac{1}{2} t^\top K_I t\right) \Phi_2\left(D_I^{-1}(m_I + K_I t); R_I\right). \tag{19}$$

Then

$$[\mathfrak{Q}_\rho(m, K)]_{ij} = \frac{\partial^2}{\partial t_1 \partial t_2} \Psi_I(t)\bigg|_{t=0}. \tag{20}$$

Equation (20) follows by exponential tilting: the factor in (19) is

$$\mathbb{E}\left[e^{t^\top G_I} \mathbf{1}_{\{G_i > 0, G_j > 0\}}\right], \tag{21}$$

so differentiation inserts $G_i G_j$. Thus all first and second moments required by the algorithm are exact and reducible to univariate and bivariate Gaussian special functions; no Monte Carlo is needed.

4 ReLU operator: affine preactivation state to nonlinear postactivation state

Suppose the incoming preactivation state is

$$Y \sim \nu_Y, \qquad C \mid Y = y \rightsquigarrow \mathcal{N}(\mu_0 + \mu_1 y, \Sigma_0 + \Sigma_1 y). \tag{22}$$

Set

$$A = \rho(Y), \qquad B = \rho(C). \tag{23}$$

For every scalar slice at which the covariance is valid, define

$$r(y) = \mathfrak{m}_\rho(\mu(y), \Sigma(y)), \qquad R(y) = \mathfrak{S}_\rho(\mu(y), \Sigma(y)). \tag{24}$$

These are generally nonlinear functions of y, even though $\mu(y)$ and $\Sigma(y)$ are affine.

4.1 Positive branch

Proposition 4.1 (Positive ReLU slices)**.** *For every $a > 0$,*

$$\mathcal{L}(B \mid A = a) = \mathcal{L}(\rho(C) \mid Y = a) \tag{25}$$

under the surrogate (22)*. Consequently,*

$$m^+(a) = \mathbb{E}[B \mid A = a] = r(a), \qquad S^+(a) = \mathrm{Cov}(B \mid A = a) = R(a). \tag{26}$$

Proof. For $a > 0$, the event $A = a$ is equivalent to $Y = a$ because $A = \max\{Y, 0\}$. Conditional on $Y = a$, the affine preactivation model gives a Gaussian bulk slice, and section 3 computes the first two moments of its coordinatewise ReLU exactly. □

In coordinates, if

$$\mu_i(a) = (\mu_0)_i + (\mu_1)_i a, \qquad \sigma_i(a)^2 = (\Sigma_0)_{ii} + (\Sigma_1)_{ii} a, \tag{27}$$

then

$$m_i^+(a) = \mu_i(a)\Phi\left(\frac{\mu_i(a)}{\sigma_i(a)}\right) + \sigma_i(a)\phi\left(\frac{\mu_i(a)}{\sigma_i(a)}\right). \tag{28}$$

This explicitly exhibits the nonlinear dependence generated by ReLU.

4.2 The zero atom

All nonpositive values of Y map to the same scalar output $A = 0$. Let

$$p_0^+ = \mathbb{P}(Y \le 0) = \nu_Y((-\infty, 0]). \tag{29}$$

When $p_0^+ > 0$, define the normalized negative-branch measure

$$\nu_Y^-(\mathrm{d}y) = \frac{\mathbf{1}_{\{y \le 0\}} \nu_Y(\mathrm{d}y)}{p_0^+}. \tag{30}$$

Proposition 4.2 (Exact recombination at the zero atom)**.** *Under the affine Gaussian surrogate* (22)*, the first two conditional moments of the bulk at the ReLU atom are*

$$m_0^+ = \mathbb{E}[B \mid A = 0] = \int r(y)\, \nu_Y^-(\,\mathrm{d}y), \tag{31}$$

$$S_0^+ = \int \Big[R(y) + (r(y) - m_0^+)(r(y) - m_0^+)^\top\Big]\, \nu_Y^-(\,\mathrm{d}y). \tag{32}$$

Proof. Since $A = 0$ if and only if $Y \le 0$, (31) is the law of total expectation over the latent preactivation value Y. Equation (32) is the law of total covariance: $R(y)$ is the within-slice covariance of $\rho(C)$ and the outer product is the between-slice covariance of the conditional means. □

It is essential to apply the Gaussian–ReLU map separately at each negative slice and only then merge the resulting first two moments. In general,

$$\mathbb{E}[\rho(C) \mid Y \le 0] \ne \mathbb{E}_{G\sim\mathcal{N}(\mathbb{E}[C|Y\le 0],\mathrm{Cov}(C|Y\le 0))}[\rho(G)]. \tag{33}$$

Merging the preactivation mixture first would introduce an additional, unnecessary Gaussianization error.

4.3 Output scalar law

The post-ReLU spike law is

$$\nu^+ = p_0^+\delta_0 + \nu_Y \restriction_{(0,\infty)}\,. \tag{34}$$

Together, (26), (31), and (32) define the nonlinear post-ReLU state

$$\mathsf{S}^+ = (\nu^+, m^+, S^+). \tag{35}$$

4.4 Finite-grid ReLU update

Suppose the signed scalar law is represented by nodes and weights

$$\nu_Y^{(m)} = \sum_{j=1}^{m} w_j \delta_{y_j}, \qquad \sum_j w_j = 1, \tag{36}$$

and suppose an edge is placed at zero so that no cell straddles the ReLU kink. For each node compute

$$r_j = \mathfrak{m}_\rho(\mu_0 + \mu_1 y_j, \Sigma_0 + \Sigma_1 y_j), \tag{37}$$

$$R_j = \mathfrak{S}_\rho(\mu_0 + \mu_1 y_j, \Sigma_0 + \Sigma_1 y_j). \tag{38}$$

Every $y_j > 0$ remains a separate post-ReLU node

$$a_j^+ = y_j, \qquad p_j^+ = w_j, \qquad m_j^+ = r_j, \qquad S_j^+ = R_j. \tag{39}$$

All nodes with $y_j \le 0$ merge into

$$p_0^+ = \sum_{j:y_j\le 0} w_j, \tag{40}$$

$$m_0^+ = \frac{1}{p_0^+} \sum_{j:y_j\le 0} w_j r_j, \tag{41}$$

$$S_0^+ = \frac{1}{p_0^+} \sum_{j:y_j\le 0} w_j \Big[R_j + (r_j - m_0^+)(r_j - m_0^+)^\top\Big]. \tag{42}$$

These are the exact discrete analogues of (31)–(32).

5 Linear operator: nonlinear postactivation state to reconditioned preactivation moments

We now begin from a post-ReLU state

$$A \sim \nu, \qquad B \mid A = a \rightsquigarrow \mathcal{N}(m(a), S(a)), \tag{43}$$

where $m(a)$ and $S(a)$ may be nonlinear. Apply the block linear map (5).

5.1 Conditional joint Gaussian for a fixed old spike value

Define

$$y(a) = \gamma a + r^\top m(a) + \beta, \tag{44}$$
$$c(a) = ua + Vm(a) + \eta, \tag{45}$$
$$q(a) = r^\top S(a) r, \tag{46}$$
$$g(a) = VS(a) r, \tag{47}$$
$$T(a) = VS(a)V^\top. \tag{48}$$

Proposition 5.1 (Conditional linear transformation)**.** *Under the conditional Gaussian closure* (43)*,*

$$\begin{pmatrix} Y \\ C \end{pmatrix} \mid A = a \sim \mathcal{N}\left(\begin{pmatrix} y(a) \\ c(a) \end{pmatrix}, \begin{pmatrix} q(a) & g(a)^\top \\ g(a) & T(a) \end{pmatrix}\right). \tag{49}$$

Proof. Conditional on $A = a$, both Y and C are affine functions of the Gaussian surrogate $B \sim \mathcal{N}(m(a), S(a))$. Their means and covariance blocks follow by direct multiplication. □

The new scalar preactivation law is therefore

$$\nu_Y(\mathrm{d}y) = \int \mathcal{N}(\mathrm{d}y; y(a), q(a))\, \nu(\mathrm{d}a). \tag{50}$$

For an atomic old law $\nu = \sum_i p_i \delta_{a_i}$,

$$\nu_Y = \sum_i p_i \mathcal{N}(y_i, q_i), \qquad y_i = y(a_i), \quad q_i = q(a_i). \tag{51}$$

5.2 Exact pointwise reconditioning under the surrogate

Assume first that $q(a) > 0$. Gaussian regression in (49) gives

$$C \mid A = a, Y = y \sim \mathcal{N}(\ell(a, y), R(a)), \tag{52}$$

where

$$\ell(a, y) = c(a) + \frac{g(a)}{q(a)}(y - y(a)), \tag{53}$$
$$R(a) = T(a) - \frac{g(a)g(a)^\top}{q(a)}. \tag{54}$$

Let $k_a(y)$ denote the density of $\mathcal{N}(y(a), q(a))$, and let

$$f_Y(y) = \int k_a(y)\, \nu(\mathrm{d}a). \tag{55}$$

At every y with $f_Y(y) > 0$, Bayes' rule gives the posterior measure

$$\pi_y(\,\mathrm{d}a) = \frac{k_a(y)\nu(\,\mathrm{d}a)}{f_Y(y)}. \tag{56}$$

Proposition 5.2 (Reconditioned first two moments)**.** *Under the incoming conditional-Gaussian surrogate, the exact conditional first two moments of the transformed bulk given the new spike value are*

$$\widehat{m}(y) = \mathbb{E}[C \mid Y = y] = \int \ell(a, y)\, \pi_y(\,\mathrm{d}a), \tag{57}$$

$$\widehat{S}(y) = \int \Big[R(a) + (\ell(a, y) - \widehat{m}(y))(\ell(a, y) - \widehat{m}(y))^\top\Big]\, \pi_y(\,\mathrm{d}a). \tag{58}$$

Proof. Equation (57) is total expectation over the posterior old spike value A. Equation (58) is total covariance, with $R(a)$ the within-component covariance and the outer product the between-component contribution. □

Although every component in (52) is Gaussian, the mixture $C \mid Y = y$ is generally not Gaussian. Retaining only (57)–(58) and replacing the mixture by one Gaussian is a conditional $K = 2$ closure.

When $q(a) = 0$, the scalar Y is the deterministic value $y(a)$ in that component. Such a branch is treated as a Dirac mass in (50); no division by $q(a)$ is performed.

5.3 Closed-form finite-cell reconditioning

The pointwise formulas are ideal for adaptive quadrature. A finite-grid implementation can instead condition on scalar cells, with all required integrals available in closed form.

Suppose the incoming post-ReLU state is atomic,

$$\nu = \sum_{i=1}^{m} p_i \delta_{a_i}, \qquad B \mid A = a_i \rightsquigarrow \mathcal{N}(m_i, S_i). \tag{59}$$

For component i, abbreviate

$$m_{Y,i} = \gamma a_i + r^\top m_i + \beta, \qquad s_{Y,i}^2 = r^\top S_i r, \tag{60}$$

$$m_{C,i} = u a_i + V m_i + \eta, \qquad g_i = V S_i r, \qquad S_{C,i} = V S_i V^\top. \tag{61}$$

Let the signed preactivation grid consist of cells

$$I_j = [\ell_j, h_j), \qquad -\infty = \ell_1 < h_1 = \ell_2 < \cdots < h_J = +\infty, \tag{62}$$

with zero included as an edge.

For $s_{Y,i} > 0$, set

$$\alpha_{ji} = \frac{\ell_j - m_{Y,i}}{s_{Y,i}}, \qquad \beta_{ji} = \frac{h_j - m_{Y,i}}{s_{Y,i}}, \qquad Q_{ji} = \Phi(\beta_{ji}) - \Phi(\alpha_{ji}). \tag{63}$$

For $Q_{ji} > 0$, define

$$\delta_{ji} = s_{Y,i} \frac{\phi(\alpha_{ji}) - \phi(\beta_{ji})}{Q_{ji}}, \tag{64}$$

$$v_{ji} = s_{Y,i}^2 \left[1 + \frac{\alpha_{ji}\phi(\alpha_{ji}) - \beta_{ji}\phi(\beta_{ji})}{Q_{ji}}\right] - \delta_{ji}^2. \tag{65}$$

Then

$$y_{i\to j} = \mathbb{E}[Y \mid A = a_i, Y \in I_j] = m_{Y,i} + \delta_{ji}, \tag{66}$$

$$m_{i\to j} = \mathbb{E}[C \mid A = a_i, Y \in I_j] = m_{C,i} + \frac{g_i}{s_{Y,i}^2}\delta_{ji}, \tag{67}$$

$$S_{i\to j} = S_{C,i} - \frac{g_i g_i^\top}{s_{Y,i}^2} + \frac{g_i g_i^\top}{s_{Y,i}^4} v_{ji}. \tag{68}$$

These formulas are exact consequences of Gaussian regression and truncated-normal moments.

The total cell mass and posterior component weights are

$$w_j = \sum_i p_i Q_{ji}, \qquad \eta_{i|j} = \frac{p_i Q_{ji}}{w_j}. \tag{69}$$

The scalar cell representative is its exact mixture centroid,

$$y_j = \sum_i \eta_{i|j} y_{i\to j}. \tag{70}$$

The cell-conditioned bulk moments are

$$\widehat{m}_j = \sum_i \eta_{i|j} m_{i\to j}, \tag{71}$$

$$\widehat{S}_j = \sum_i \eta_{i|j} \left[S_{i\to j} + (m_{i\to j} - \widehat{m}_j)(m_{i\to j} - \widehat{m}_j)^\top \right]. \tag{72}$$

Thus $(w_j, y_j, \widehat{m}_j, \widehat{S}_j)$ is an exact finite-cell summary under the incoming $K = 2$ surrogate. The only scalar approximation is collapsing the within-cell variation of Y to the representative y_j.

If $s_{Y,i} = 0$, component i is assigned wholly to the unique cell containing $m_{Y,i}$, with

$$y_{i\to j} = m_{Y,i}, \qquad m_{i\to j} = m_{C,i}, \qquad S_{i\to j} = S_{C,i}. \tag{73}$$

6 Affine re-projection after linear reconditioning

The functions $\widehat{m}(y)$ and $\widehat{S}(y)$ from (57)–(58) are generally nonlinear because the posterior π_y changes nonlinearly with y. The defining approximation of the analytic method is to project these functions onto affine dependence on the new spike coordinate.

6.1 Best affine approximation of the conditional mean

Let

$$\bar{y} = \mathbb{E}[Y], \qquad v_Y = \mathrm{Var}(Y). \tag{74}$$

Assume $v_Y > 0$. Define

$$(\mu_0, \mu_1) = \operatorname*{arg\,min}_{a_0, a_1 \in \mathbb{R}^d} \int \|\widehat{m}(y) - a_0 - a_1 y\|_2^2 \, \nu_Y(\,\mathrm{d}y). \tag{75}$$

Proposition 6.1 (Mean projection)**.** *The unique minimizer of* (75) *is*

$$\mu_1 = \frac{\mathbb{E}[(Y - \bar{y})\widehat{m}(Y)]}{v_Y} = \frac{\mathrm{Cov}(Y, C)}{\mathrm{Var}(Y)}, \tag{76}$$

$$\mu_0 = \mathbb{E}[\widehat{m}(Y)] - \mu_1 \bar{y} = \mathbb{E}[C] - \mu_1 \mathbb{E}[Y]. \tag{77}$$

The residual

$$e_m(y) = \widehat{m}(y) - \mu_0 - \mu_1 y \tag{78}$$

satisfies

$$\mathbb{E}[e_m(Y)] = 0, \qquad \mathbb{E}[(Y - \bar{y})e_m(Y)] = 0. \tag{79}$$

Proof. This is orthogonal projection in the Hilbert space $L^2(\nu_Y; \mathbb{R}^d)$ onto the span of 1 and y. Differentiating the quadratic objective with respect to a_0 and a_1 gives the normal equations (79), whose solution is (76)–(77). The identity $\mathbb{E}[(Y - \bar{y})\widehat{m}(Y)] = \mathrm{Cov}(Y, C)$ follows from iterated expectation. □

If $v_Y = 0$, the slope is unidentifiable; the canonical choice is $\mu_1 = 0$ and $\mu_0 = \mathbb{E}[C]$.

6.2 Best affine approximation of the conditional covariance

The literal functional projection is

$$(\Sigma_0^{\mathrm{LS}}, \Sigma_1^{\mathrm{LS}}) = \operatorname*{arg\,min}_{A_0, A_1 \in \mathrm{Sym}_d} \int \|\widehat{S}(y) - A_0 - A_1 y\|_F^2 \, \nu_Y(\,\mathrm{d}y). \tag{80}$$

Proposition 6.2 (Unconstrained covariance projection)**.** *If $v_Y > 0$, the unconstrained minimizer of* (80) *is*

$$\Sigma_1^{\mathrm{LS}} = \frac{\mathbb{E}[(Y - \bar{y})\widehat{S}(Y)]}{v_Y}, \tag{81}$$

$$\Sigma_0^{\mathrm{LS}} = \mathbb{E}[\widehat{S}(Y)] - \bar{y}\Sigma_1^{\mathrm{LS}}. \tag{82}$$

The matrix residual

$$e_S(y) = \widehat{S}(y) - \Sigma_0^{\mathrm{LS}} - y\Sigma_1^{\mathrm{LS}} \tag{83}$$

satisfies the entrywise orthogonality conditions

$$\mathbb{E}[e_S(Y)] = 0, \qquad \mathbb{E}[(Y - \bar{y})e_S(Y)] = 0. \tag{84}$$

Proof. Apply scalar least squares to every matrix entry. Symmetry is preserved because $\widehat{S}(y)$ is symmetric. □

For the finite-grid state $(w_j, y_j, \widehat{m}_j, \widehat{S}_j)$, the formulas become

$$\bar{y} = \sum_j w_j y_j, \qquad v_Y = \sum_j w_j (y_j - \bar{y})^2, \tag{85}$$

$$\mu_1 = \frac{\sum_j w_j (y_j - \bar{y})\widehat{m}_j}{v_Y}, \qquad \mu_0 = \sum_j w_j \widehat{m}_j - \mu_1 \bar{y}, \tag{86}$$

$$\Sigma_1^{\mathrm{LS}} = \frac{\sum_j w_j (y_j - \bar{y})\widehat{S}_j}{v_Y}, \qquad \Sigma_0^{\mathrm{LS}} = \sum_j w_j \widehat{S}_j - \bar{y}\Sigma_1^{\mathrm{LS}}. \tag{87}$$

6.3 Optional moment-conservative covariance correction

The literal least-squares projection minimizes the error of the conditional covariance function, but the nonlinear mean residual contributes to the unconditional bulk covariance. Define

$$\mathcal{R}_m = \mathbb{E}[e_m(Y)e_m(Y)^\top] \succeq 0. \tag{88}$$

By (79),

$$\mathrm{Cov}(\widehat{m}(Y)) = v_Y \mu_1 \mu_1^\top + \mathcal{R}_m. \tag{89}$$

If one uses $(\Sigma_0^{\mathrm{LS}}, \Sigma_1^{\mathrm{LS}})$ unchanged, the surrogate unconditional covariance of C is smaller than the covariance implied by $(\widehat{m}, \widehat{S})$ by exactly $\mathcal{R}_m$.

A moment-conservative option is therefore

$$\Sigma_1^{\mathrm{MC}} = \Sigma_1^{\mathrm{LS}}, \qquad \Sigma_0^{\mathrm{MC}} = \Sigma_0^{\mathrm{LS}} + \mathcal{R}_m. \tag{90}$$

Then

$$\mathbb{E}[\Sigma_0^{\mathrm{MC}} + Y\Sigma_1^{\mathrm{MC}}] + \mathrm{Cov}(\mu_0 + \mu_1 Y) = \mathbb{E}[\widehat{S}(Y)] + \mathrm{Cov}(\widehat{m}(Y)) = \mathrm{Cov}(C). \tag{91}$$

Thus the correction preserves the full unconditional bulk covariance while retaining the best affine mean and the best covariance slope. The literal functional algorithm uses the LS coefficients; a K-prop implementation may prefer the MC intercept because global second moments are primary tracked quantities.

6.4 Positive-semidefinite covariance constraints

An unconstrained affine fit need not be positive semidefinite at every scalar value where it is evaluated. On a finite grid, the rigorous repair is the convex semidefinite least-squares problem

$$\min_{A_0, A_1 \in \mathrm{Sym}_d} \sum_j w_j \|\widehat{S}_j - A_0 - y_j A_1\|_F^2 \tag{92}$$

subject to

$$A_0 + y_j A_1 \succeq \varepsilon I_d \qquad \text{for every retained node } y_j, \tag{93}$$

where $\varepsilon \geq 0$ is a numerical floor. If the unconstrained coefficients satisfy the constraints, they are already the constrained optimum.

The following obstruction explains why a finite grid, clipping, or another PSD parameterization is necessary for a nontrivial continuous signed model.

Proposition 6.3 (Global PSD obstruction)**.** *If $\Sigma_0 + y\Sigma_1 \succeq 0$ for every $y \in \mathbb{R}$, then $\Sigma_1 = 0$.*

Proof. For every $v \in \mathbb{R}^d$, the scalar affine function

$$v^\top \Sigma_0 v + y\, v^\top \Sigma_1 v \tag{94}$$

must be nonnegative for all positive and negative y. Hence $v^\top \Sigma_1 v = 0$ for every v, which implies $\Sigma_1 = 0$. □

A continuous backend can instead use

$$\Sigma(y) = \Sigma_0 + \Sigma_1 \,\mathrm{clip}(y, -T, T) \tag{95}$$

or a PSD factorization such as

$$\Sigma(y) = L(y)L(y)^\top, \qquad L(y) = L_0 + L_1 \,\mathrm{clip}(y, -T, T). \tag{96}$$

The finite-grid algorithm below keeps the exact affine notation and enforces (93).

7 Complete propagation algorithms

7.1 Ideal continuous-slice operator

Algorithm 7.1 (Continuous analytic layer update)**.** Given a post-ReLU state (ν, m, S) and a layer block $(\gamma, r, u, V, \beta, \eta)$:

1. **Conditional $K = 2$ closure.** Replace $B \mid A = a$ by $\mathcal{N}(m(a), S(a))$.
2. **Linear transformation.** Compute $y(a), c(a), q(a), g(a), T(a)$ from (44)–(48).
3. **New scalar law.** Form the Gaussian mixture ν_Y in (50).
4. **Recondition on the new spike.** Compute $\widehat{m}(y)$ and $\widehat{S}(y)$ by (57)–(58).
5. **Affine re-projection.** Compute (μ_0, μ_1) and (Σ_0, Σ_1) by (76)–(82), optionally apply (90), and enforce a valid covariance parameterization.
6. **Slice-wise ReLU.** Compute $r(y), R(y)$ by (24).
7. **Zero atom.** Merge every $y \leq 0$ using (29)–(32).
8. **Positive branch.** For every $a > 0$, retain $m^+(a) = r(a)$ and $S^+(a) = R(a)$ without a new affine fit.
9. **Return** (ν^+, m^+, S^+) with scalar law (34).

The only obstacle to implementing Algorithm 7.1 literally with $\Sigma_0 + y\Sigma_1$ is global PSD validity. Adaptive quadrature on a bounded high-probability interval plus clipping outside that interval is one rigorous solution.

7.2 Recommended finite-grid algorithm

Algorithm 7.2 (Finite-grid analytic affine $K = 2$ propagation)**.** Maintain the post-ReLU arrays

$$\{p_i, a_i, m_i, S_i\}_{i=0}^{m-1}, \qquad a_i \geq 0, \quad S_i \succeq 0, \tag{97}$$

with a dedicated $a_0 = 0$ state when needed. For each hidden layer:

1. Choose signed preactivation cells $I_j = [\ell_j, h_j)$ with an edge exactly at zero.
2. For every old node i, compute the component parameters (60)–(61).
3. For every pair (j, i), compute $Q_{ji}, \delta_{ji}, v_{ji}$ and the exact cell-conditioned moments (66)–(68).
4. Merge old components within each new cell using (69)–(72), obtaining $(w_j, y_j, \widehat{m}_j, \widehat{S}_j)$.
5. Fit the global affine conditional mean by (86).
6. Fit the affine conditional covariance by (87); optionally add the moment-conservative intercept (90); enforce (93) if necessary.
7. At every signed node y_j, form

   $$\widetilde{m}_j = \mu_0 + \mu_1 y_j, \qquad \widetilde{S}_j = \Sigma_0 + \Sigma_1 y_j, \tag{98}$$

   and compute the exact Gaussian–ReLU moments

   $$r_j = \mathfrak{m}_\rho(\widetilde{m}_j, \widetilde{S}_j), \qquad R_j = \mathfrak{S}_\rho(\widetilde{m}_j, \widetilde{S}_j). \tag{99}$$

8. Keep every positive node by (39). Merge all nonpositive nodes into the zero state by (40)–(42).

9. Optionally re-quantize only the positive scalar nodes to a fixed budget, merging bulk moments by total expectation and total covariance.

The finite-grid state returned by Step 8 again contains nonlinear dependence of (m_i, S_i) on the positive scalar values a_i. No affine fit is made until after the next linear reconditioning.

7.3 Scalar-law variants to compare experimentally

The same bulk update supports three scalar backends.

A. Exact Gaussian-mixture cells. Keep the mixture (51) long enough to compute exact cell masses, centroids, and truncated moments. This is the recommended accurate backend.

B. Atomic quadrature. Approximate the mixture immediately by nodes and weights, evaluate the pointwise posterior formulas at those nodes, and continue with finite sums. This is cheaper but adds quadrature error.

C. Single moment-matched Gaussian. Replace the scalar mixture by one Gaussian before ReLU. This is a useful baseline but introduces a stronger scalar modeling error and should not be the default.

At equal bulk closure and equal scalar resolution, Backend A has no larger scalar-law approximation than a quantized or moment-matched replacement. Nevertheless, downstream output error need not be monotone because the affine fit and subsequent nonlinearities can interact with numerical resolution; an ablation is therefore required.

8 Worked one-layer walkthrough

This example makes the nonlinear reconditioning and the location of the affine fit explicit. Let the old spike have two atoms

$$\mathbb{P}(A = a_i) = p_i, \qquad i \in \{0, 1\}, \qquad a_0 = 0, \quad a_1 = s > 0, \quad p_0 + p_1 = 1. \tag{100}$$

Let the old bulk be two-dimensional and conditionally independent,

$$B = (B_1, B_2), \qquad B \mid A = a_i \rightsquigarrow \mathcal{N}\left(\begin{pmatrix} m_{i1} \\ m_{i2} \end{pmatrix}, \begin{pmatrix} \sigma_{i1}^2 & 0 \\ 0 & \sigma_{i2}^2 \end{pmatrix}\right). \tag{101}$$

Consider a linear transformation with scalar new spike and scalar new bulk

$$Y = \gamma A + B_1, \qquad C = uA + \lambda B_1 + B_2. \tag{102}$$

For old state i,

$$y_i = \gamma a_i + m_{i1}, \qquad q_i = \sigma_{i1}^2, \tag{103}$$

$$c_i = u a_i + \lambda m_{i1} + m_{i2}, \qquad g_i = \lambda \sigma_{i1}^2, \tag{104}$$

$$T_i = \lambda^2 \sigma_{i1}^2 + \sigma_{i2}^2. \tag{105}$$

Hence

$$Y \mid A = a_i \sim \mathcal{N}(y_i, \sigma_{i1}^2), \tag{106}$$

and Gaussian regression yields

$$C \mid A = a_i, Y = y \sim \mathcal{N}\big(c_i + \lambda(y - y_i), \sigma_{i2}^2\big). \tag{107}$$

The posterior old-state weights are

$$\eta_i(y) = \frac{p_i \sigma_{i1}^{-1} \phi\big((y - y_i)/\sigma_{i1}\big)}{\sum_{k=0}^{1} p_k \sigma_{k1}^{-1} \phi\big((y - y_k)/\sigma_{k1}\big)}. \tag{108}$$

Therefore

$$\widehat{m}(y) = \sum_{i=0}^{1} \eta_i(y)\big[c_i + \lambda(y - y_i)\big], \tag{109}$$

$$\widehat{S}(y) = \sum_{i=0}^{1} \eta_i(y) \left[\sigma_{i2}^2 + \big(c_i + \lambda(y - y_i) - \widehat{m}(y)\big)^2\right]. \tag{110}$$

Even though each component mean in (107) is affine in y, the posterior weights (108) are nonlinear. Thus both (109) and (110) are generally nonlinear. This is the exact point at which the affine re-projection is needed.

Let f_Y be the two-component mixture density. Compute

$$\bar{y} = \int y f_Y(y)\,\mathrm{d}y, \qquad v_Y = \int (y - \bar{y})^2 f_Y(y)\,\mathrm{d}y, \tag{111}$$

then fit

$$\mu_1 = \frac{\int (y - \bar{y})\widehat{m}(y) f_Y(y)\,\mathrm{d}y}{v_Y}, \qquad \mu_0 = \int \widehat{m}(y) f_Y(y)\,\mathrm{d}y - \mu_1 \bar{y}, \tag{112}$$

$$\Sigma_1 = \frac{\int (y - \bar{y})\widehat{S}(y) f_Y(y)\,\mathrm{d}y}{v_Y}, \qquad \Sigma_0 = \int \widehat{S}(y) f_Y(y)\,\mathrm{d}y - \Sigma_1 \bar{y}. \tag{113}$$

At a positive preactivation value $y > 0$, the fitted slice is

$$C \mid Y = y \rightsquigarrow \mathcal{N}(\mu(y), \sigma(y)^2), \qquad \mu(y) = \mu_0 + \mu_1 y, \quad \sigma(y)^2 = \Sigma_0 + \Sigma_1 y. \tag{114}$$

The next post-ReLU bulk moments are therefore

$$m^+(y) = \mu(y)\Phi\left(\frac{\mu(y)}{\sigma(y)}\right) + \sigma(y)\phi\left(\frac{\mu(y)}{\sigma(y)}\right), \tag{115}$$

$$q^+(y) = \big(\mu(y)^2 + \sigma(y)^2\big)\Phi\left(\frac{\mu(y)}{\sigma(y)}\right) + \mu(y)\sigma(y)\phi\left(\frac{\mu(y)}{\sigma(y)}\right), \tag{116}$$

$$S^+(y) = q^+(y) - m^+(y)^2. \tag{117}$$

For the zero atom,

$$p_0^+ = \int_{-\infty}^{0} f_Y(y)\,\mathrm{d}y, \tag{118}$$

$$m_0^+ = \frac{1}{p_0^+} \int_{-\infty}^{0} m^+(y) f_Y(y)\,\mathrm{d}y, \tag{119}$$

$$S_0^+ = \frac{1}{p_0^+} \int_{-\infty}^{0} \left[S^+(y) + (m^+(y) - m_0^+)^2\right] f_Y(y)\,\mathrm{d}y. \tag{120}$$

The positive functions (115) and (117) are retained as nonlinear dependence for the next layer.

9 Initialization and readout

Suppose

$$X^{(0)} \sim \mathcal{N}(0, I_n). \tag{121}$$

Then

$$A^{(0)} = e^\top X^{(0)} \sim \mathcal{N}(0, 1), \qquad B^{(0)} = \Pi X^{(0)} \sim \mathcal{N}(0, I_d), \tag{122}$$

and the two are independent. Thus the exact initial slice state is

$$\nu_0 = \mathcal{N}(0, 1), \qquad m_0(a) = 0, \qquad S_0(a) = I_d. \tag{123}$$

Although the initial spike is signed, the linear formulas in section 5 apply without change. After the first ReLU, the state has support on $[0, \infty)$ and a zero atom.

For a final linear readout

$$f(X) = w_A A + w_B^\top B + b, \tag{124}$$

the continuous-state prediction is

$$\mathbb{E}[f(X)] = w_A \int a \, \nu(\,\mathrm{d}a) + w_B^\top \int m(a) \, \nu(\,\mathrm{d}a) + b. \tag{125}$$

For the finite state,

$$\mathbb{E}[f(X)] = w_A \sum_i p_i a_i + w_B^\top \sum_i p_i m_i + b. \tag{126}$$

The unconditional covariance, when required, follows from

$$\mathrm{Cov}(B) = \int \Big[S(a) + (m(a) - \bar{m})(m(a) - \bar{m})^\top \Big] \, \nu(\,\mathrm{d}a), \qquad \bar{m} = \int m(a) \nu(\,\mathrm{d}a). \tag{127}$$

10 Exactness and error decomposition

10.1 What is exact under the current surrogate

Theorem 10.1 (One-layer accounting)**.** *Fix a post-ReLU state (ν, m, S) and suppose the conditional-Gaussian closure*

$$B \mid A = a \rightsquigarrow \mathcal{N}(m(a), S(a)) \tag{128}$$

is adopted. Then:

(i) *the conditional joint law (49) and scalar mixture (50) are exact under that surrogate;*

(ii) *the reconditioned functions (57)–(58), or their finite-cell counterparts (71)–(72), are exact first two moments under that surrogate;*

(iii) *the affine coefficients are the unique L^2-optimal coefficients for the stated projection objective, subject to any explicitly imposed PSD constraints;*

(iv) *the Gaussian–ReLU first two moments at every retained slice are exact;*

(v) *the zero-atom merge is exact at the level of first two moments under the affine Gaussian preactivation surrogate.*

Consequently, with exact scalar integration, the only conceptual approximations are conditional $K = 2$ Gaussian closure and affine re-projection. A finite scalar grid adds one-dimensional discretization.

Proof. Items (i) and (ii) are section 5; item (iii) is section 6; item (iv) is section 3; item (v) is section 4. No other approximation is introduced by the displayed formulas. □

10.2 Conditional higher-cumulant truncation

There are two places where a conditional law is generally non-Gaussian but is represented only through its first two moments.

1. After ReLU, $\rho(C) \mid Y = y$ is a rectified Gaussian and therefore non-Gaussian except in degenerate cases. Its mean and covariance are retained exactly, while conditional cumulants of order $r \geq 3$ are discarded before the next linear layer.
2. After linear reconditioning, $C \mid Y = y$ is generally a mixture over the old spike value A. Equations (57)–(58) retain its exact first two moments under the incoming surrogate, after which the mixture is replaced by a Gaussian.

These are both instances of the same conditional $K = 2$ closure. If the relevant conditional moment generating functions exist near the origin, the omitted information is equivalently the tower of conditional cumulants

$$\kappa_r(B \mid A = a), \qquad r \geq 3. \tag{129}$$

10.3 Affine re-projection error

The nonlinear conditional moment functions after linear reconditioning are replaced by affine functions. The intrinsic residuals are

$$\mathcal{E}_m = \int \|e_m(y)\|_2^2 \, \nu_Y(\,\mathrm{d}y), \tag{130}$$

$$\mathcal{E}_S = \int \|e_S(y)\|_F^2 \, \nu_Y(\,\mathrm{d}y). \tag{131}$$

For the finite grid, replace the integrals by weighted sums. These quantities should be logged at every layer. They directly test the modeling hypothesis that conditional mean and covariance are approximately affine in the new spike coordinate.

The mean projection also has an interpretable matrix residual

$$\mathcal{R}_m = \mathbb{E}[e_m(Y)e_m(Y)^\top], \tag{132}$$

which is exactly the unconditional bulk covariance lost by the literal LS surrogate and restored by the moment-conservative correction (90).

10.4 Scalar discretization error

If the scalar law is replaced by an atomic approximation

$$\nu \approx \sum_i p_i \delta_{a_i}, \tag{133}$$

there is an additional one-dimensional quadrature or quantization error. For interval cells with centroid representatives, the squared within-cell scalar distortion is

$$\sum_i \int_{I_i} (a - a_i)^2 \nu(\,\mathrm{d}a). \tag{134}$$

This term vanishes in the ideal continuous backend and can be controlled independently of the bulk closure by increasing scalar resolution or using adaptive mixture quadrature.

10.5 No hidden ReLU moment approximation

Once a slice has been replaced by a valid Gaussian $\mathcal{N}(\mu(y), \Sigma(y))$, the first two moments after ReLU are computed exactly by section 3. Therefore the ReLU step does not introduce an additional first-two-moment approximation beyond the preceding Gaussian closure and affine fit. Likewise, the zero atom is recombined exactly by total expectation and total covariance.

11 Backend comparison and validation protocol

The choice between a continuous rectified Gaussian-mixture scalar law and a finite discrete scalar law is empirical rather than purely formal. The following ablation keeps the bulk model fixed and isolates the scalar representation.

11.1 Variants

V1. Mixture-integral backend: retain the exact Gaussian mixture after each linear step, use adaptive one-dimensional integration for the affine projection and zero atom, and use clipping or a bounded fit for covariance validity.

V2. Exact-cell backend: use the closed-form cell formulas in section 5.3, followed by the finite-grid affine fit and exact discrete zero merge.

V3. Atomic-node backend: quantize the scalar mixture immediately and use pointwise posterior weights at the nodes.

V4. Single-Gaussian baseline: moment-match the scalar mixture to one Gaussian before ReLU.

11.2 Quantities to record

For each width, depth, and network seed, record

1. relative error of the predicted final output mean against high-precision Monte Carlo;
2. scalar Wasserstein or cell distortion (134);
3. layerwise affine residuals $\mathcal{E}_m$ and $\mathcal{E}_S$;
4. the covariance residual $\mathrm{tr}(\mathcal{R}_m)$;
5. the minimum eigenvalue of every fitted slice covariance and the size of any PSD correction;
6. runtime and memory.

To compare scalar backends fairly, use the same Gaussian–ReLU kernel, the same affine projection convention, the same bulk matrix representation, and comparable numbers of scalar function evaluations. The mixture-integral backend removes scalar quantization as a conceptual error, but the finite exact-cell backend may be more stable and faster; downstream output error must be measured rather than inferred solely from scalar-law fidelity.

12 Implementation checklist

A correct implementation should satisfy the following invariants at every layer.

1. All scalar masses are nonnegative and sum to one up to a logged numerical tolerance.
2. The signed preactivation grid has an edge exactly at zero.
3. Every covariance supplied to the Gaussian–ReLU kernel is symmetric and PSD at the evaluation node.
4. The zero state is formed after slice-wise ReLU, not before.
5. Every mixture merge includes the between-component outer-product term.
6. The affine fit is performed after linear reconditioning and before ReLU.
7. No affine fit is performed on the positive post-ReLU functions $m^+(a), S^+(a)$.
8. The degenerate branch $s^2_{Y,i} = 0$ is handled without division.
9. Layer logs include $\mathcal{E}_m$, $\mathcal{E}_S$, $\mathrm{tr}(\mathcal{R}_m)$, scalar distortion, and PSD corrections.
10. The readout uses the mixture expectation (125) or (126), not a single collapsed spike value.

13 Conclusion

The rigorous analytic design is the composition

$$\boxed{\begin{array}{c}\text{nonlinear post-ReLU slice state } (\nu, m(a), S(a)) \\ \downarrow\ \text{conditional } K=2 \text{ closure and linear map} \\ \text{exact reconditioned moments } \widehat{m}(y), \widehat{S}(y) \\ \downarrow\ \text{best affine re-projection} \\ C \mid Y = y \rightsquigarrow \mathcal{N}(\mu_0 + \mu_1 y, \Sigma_0 + \Sigma_1 y) \\ \downarrow\ \text{exact Gaussian–ReLU moments} \\ \text{nonlinear positive slices plus a recombined zero atom.}\end{array}} \tag{135}$$

With exact one-dimensional integration, the model has exactly two conceptual approximation mechanisms: discarded conditional cumulants above order two, and the affine re-projection of the reconditioned first two moments. The recommended finite-grid implementation adds a controlled scalar discretization term but resolves the global PSD obstruction of a nonconstant affine covariance on an unbounded signed coordinate. This gives a precise algorithm that can be implemented, audited, and compared directly against the existing binned method.

A Truncated-normal formulas used by the finite-cell update

Let $Z \sim \mathcal{N}(0, 1)$ and condition on $\alpha \le Z < \beta$. With

$$Q = \Phi(\beta) - \Phi(\alpha), \tag{136}$$

we have

$$\mathbb{E}[Z \mid \alpha \le Z < \beta] = \frac{\phi(\alpha) - \phi(\beta)}{Q}, \tag{137}$$

$$\mathbb{E}[Z^2 \mid \alpha \le Z < \beta] = 1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Q}, \tag{138}$$

$$\operatorname{Var}(Z \mid \alpha \le Z < \beta) = 1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Q} - \left(\frac{\phi(\alpha) - \phi(\beta)}{Q}\right)^2. \tag{139}$$

For $Y = m + sZ$, multiply the conditional mean shift by s and the conditional variance by s^2. The conventions $x\phi(x) \to 0$ as $x \to \pm\infty$ handle unbounded cells.

B Derivation of the cell-conditioned bulk covariance

For a fixed old component, write

$$C = m_C + \frac{g}{s_Y^2}(Y - m_Y) + C_\perp, \tag{140}$$

where

$$C_\perp \perp Y, \qquad \operatorname{Cov}(C_\perp) = S_C - \frac{gg^\top}{s_Y^2}. \tag{141}$$

Conditioning on a cell I affects only the regression term. Hence

$$\mathbb{E}[C \mid Y \in I] = m_C + \frac{g}{s_Y^2}\mathbb{E}[Y - m_Y \mid Y \in I], \tag{142}$$

$$\begin{aligned} \operatorname{Cov}(C \mid Y \in I) &= \operatorname{Cov}(C_\perp) + \frac{gg^\top}{s_Y^4}\operatorname{Var}(Y \mid Y \in I) && (143)\\ &= S_C - \frac{gg^\top}{s_Y^2} + \frac{gg^\top}{s_Y^4}\operatorname{Var}(Y \mid Y \in I), && (144) \end{aligned}$$

which is (68).

C Relation to the companion binned construction

The companion binned method stores one Gaussian bulk law per scalar spike bin and re-Gaussianizes after every bin transition. The present method uses the same exact Gaussian regression and truncated-normal identities, but imposes a global affine structure on the reconditioned preactivation moments and then retains the nonlinear ReLU dependence until the next linear layer. The two methods therefore share the same spike–bulk split, zero-atom treatment, and total-covariance merges, while differing in where the bulk moment function is compressed.